M S ENGINEERING COLLEGE
BENGALURU

BLOCKCREDAI: DECENTRALIZED EMPLOYMENT VERIFICATION WITH AI FRAUD DETECTION

Submitted to the

Visvesvaraya Technological University, Belagavi

in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering in Information Science & Engineering

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Guide Name
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Flow of Presentation

- **1:** Abstract & Project Motivation
- **2:** Introduction & Problem Context
- **3:** Literature Review & State of the Art
- **4:** System Analysis & Problem Statement
- **5:** System Design (HLD & LLD Architecture)
- **6:** Implementation & Execution Flowchart
- **7:** Testing & Verification Matrix
- **8:** Conclusion & Future Enhancements
- **9:** Live Prototype Screenshots
- **10:** Publications & References

ABSTRACT

- **Problem Context:** Credential fabrication, inflated job titles, and fake experience certificates severely undermine hiring integrity, with global surveys indicating over 70% of resumes contain discrepancies.
- **Industry Bottleneck:** Conventional background screening relies on slow manual phone/email verifications taking 2–4 weeks, costing $50–$200 per candidate with zero cross-employer data portability.
- **Proposed Solution:** BlockCredAI delivers an end-to-end decentralized trust platform fusing EVM smart contracts with an asynchronous AI Natural Language Processing fraud detection microservice.
- **Dual-Layer Innovation:** Combines non-transferable ERC-721 Soulbound-style cryptographic badges minted by authorized employers with a bi-gram TF-IDF & Logistic Regression discrepancy classifier.
- **Key Outcomes:** Achieves sub-second on-chain verification, 0% fraud risk for cryptographic matches, automated detection of unverified claims, and zero intermediary verification fees.

INTRODUCTION

- **Background & Motivation:** The explosion of remote work and automated hiring has accelerated resume fraud; hiring teams lack a cryptographically secure method to verify historical employment claims.
- **Decentralized Identity Vision:** Transitioning from vulnerable paper certificates and forgeable PDFs to verifiable on-chain credentials cryptographically signed by authorized corporate entities.
- **Blockchain Ledger Tier:** Ethereum-compatible smart contracts store immutable employee job histories, including corporate wallet identity, job descriptions, and verified date boundaries.
- **AI Fraud Scoring Tier:** Machine learning microservice analyzes candidate resume text against claimed experiences and on-chain records, computing real-time fraud probability scores.
- **Primary Target Beneficiaries:** Recruiters (instant verification), Candidates (self-sovereign credential ownership), and Enterprise Employers (tamper-proof alumni verification).

LITERATURE REVIEW

- **1. Said et al. (IEEE Access, 2025):** A Comprehensive Blockchain-Based System for Educational Qualifications Management and Verification (ElimuChain) — Utilized Binance Smart Chain smart contracts and IPFS for decentralized credential anti-forgery validation. [Gap Addressed: Focuses strictly on static academic qualifications; lacks dynamic multi-employer tenure tracking and real-time AI/NLP resume discrepancy auditing].
- **2. Tariq et al. (IEEE TCSS, 2024):** Cerberus: A Blockchain-Based Accreditation and Degree Verification System — Designed an immutable on-chain credential verification and revocation architecture using Ethereum smart contracts to mitigate educational fraud. [Gap Addressed: Tailored for academic degree accreditation; lacks corporate tenure tracking, role transitions, or automated ML discrepancy scoring].
- **3. Sakib et al. (IEEE ICCIT, 2023):** Blockchain Based Certificate Verification for Employee Hiring & Admission System — Developed an Ethereum smart contract and Web3 wallet workflow to verify applicant credentials directly during recruitment. [Gap Addressed: Relies on manual hash matching; lacks an automated NLP engine to extract and cross-examine unstructured PDF resume claims against chain state].
- **4. Habib et al. (IEEE Access, 2022):** Online Recruitment Fraud (ORF) Detection Using Machine Learning & NLP Techniques — Evaluated TF-IDF n-grams with Random Forest and Logistic Regression classifiers, achieving high predictive accuracy on fraudulent employment text. [Gap Addressed: Analyzes isolated text patterns only; lacks cryptographic ledger integration to verify historical candidate claims against trusted issuers].
- **5. Han et al. (IEEE Access, 2022):** A Secure Academic & Employment Certificate Management System Using Blockchain Technology — Formulated a decentralized role-based access control (RBAC) model for issuing and verifying employment credentials on Ethereum. [Gap Addressed: Suffers from high public mainnet gas overhead and lacks automated semantic CV parsing to catch inflated candidate timelines].

SYSTEM ANALYSIS

- **Problems in Existing Systems:**
- **High Latency & Bottlenecks:** Manual employer HR correspondence introduces 2 to 4-week delays, stalling high-velocity recruitment pipelines.
- **Vulnerability to Forgery:** PDF offer letters, experience certificates, and reference contacts are trivial to forge using modern digital editing tools.
- **Lack of Cross-Employer Trust:** Verification performed by one firm cannot be cryptographically verified or reused by subsequent prospective employers.
- **Problem Statement:**
- **Engineering Formulation:** To design, implement, and validate a decentralized employment verification platform that registers tamper-proof cryptographic credentials on an Ethereum blockchain and integrates a machine learning NLP pipeline to detect resume fraud and tenure discrepancies in real-time.

SYSTEM DESIGN

- **Core Objectives of the Work:**
- **Cryptographic Badges:** Develop an ERC-721 smart contract to mint tamper-proof non-transferable employment badges with role-based access control.
- **Automated Discrepancy Scoring:** Build an NLP model combining TF-IDF bi-grams and logistic regression with an unverified claim penalty boost formula.
- **Multi-Tier Architectural Decoupling:** Architect cleanly separated Presentation, API Gateway, AI Microservice, and Web3 Blockchain execution layers.
- **End-to-End User Experience:** Provide a responsive Web3 dashboard allowing seamless PDF resume parsing, MetaMask signing, and instant audit verdict display.

SYSTEM DESIGN ~ HIGH LEVEL DESIGN (HLD)

BLOCKCREDAI : HIGH-LEVEL SYSTEM ARCHITECTURE (HLD)

End-to-End Multi-Tier Architecture: Presentation Tier, API Gateway, AI Intelligence, and Decentralized Blockchain

1. PRESENTATION TIER

Next.js 14 Web Portal
Responsive client interface built with React 18, TailwindCSS & Turbopack.
HTTP :3000

Web3 Wallet (MetaMask)
Injected EIP-1193 provider managing private keys, message signing, and role authentication.
Ethers.js v6

Candidate & Employer UI
PDF upload, real-time risk indicators, and on-chain credential dashboard.
Client State

2. API GATEWAY & LOGIC

Express.js REST Gateway
TypeScript controller managing security headers, rate limiting, and request validation.
PORT 4000

PDF Parser Engine
Extracts unstructured plain-text tokens from candidate resume PDFs via pdfjs-dist.
Multer Storage

Cross-Verification Matcher
Compares claimed experiences with cryptographically verified on-chain history records.
Mismatch Logic

3. AI INTELLIGENCE TIER

FastAPI Inference Engine
High-performance asynchronous microservice running ML fraud scoring.
PORT 8001

TF-IDF NLP Vectorizer
Bi-gram (1,2) text vectorization extracting 500 dominant lexical risk features.
Scikit-Learn

Logistic Regression Model
Balanced class classifier outputting base probability with mismatch risk calibration.
Joblib Model

4. BLOCKCHAIN & WEB3 TIER

Ethereum / Hardhat Node
Local EVM Cancun execution node providing JSON-RPC ledger persistence.
RPC :8545

BlockCredAI.sol Contract
ERC-721 badge minting, AccessControl role security (Admin/Employer), and job records.
Solidity 0.8.24

IPFS Storage (nft.storage)
Decentralized immutable storage for employer appointment letters and cryptographic hashes.
Content Hash

Communication Flow: Client UI → REST API Gateway → AI Model (/predict) & Blockchain RPC (eth_call / submitVerification) → NFT Badges & Fraud Scoring

BLOCKCREDAI : LOW-LEVEL DESIGN (LLD) & COMPONENT SCHEMA

Smart Contract Internal Structure, API Controller Endpoints, Data Schemas, and AI Scoring Pipeline

Smart Contract Design (BlockCredAI.sol)

Struct: JobRecord

```
+ uint256 jobId
+ address employer
+ address employee
+ string jobDetails
+ uint256 startDate, endDate
+ string ipfsHash
+ uint256 tokenId
```

Contract: BlockCredAI (ERC721, AccessControl)

```
// State Variables
- bytes32 EMPLOYER_ROLE
- mapping(address => JobRecord[]) employeeHistory
- uint256 _nextJobId, _nextTokenId

// Public Methods
+ submitVerification(employee, details, start, end, ipfs)
+ getEmployeeHistory(employee): JobRecord[]
+ verifyEmployment(employee, details): bool
+ addEmployer(employerAddress) [onlyRole(ADMIN)]
```

Backend Layer & Express Controllers

POST /api/scanResume

```
Request: FormData (walletAddress, claimedExp, resume.pdf)
Pipeline: Multer -> extractResumeText() ->
getEmployeeHistory() -> estimateMismatchCount() ->
getFraudScore()
Response: { fraudProbability, explanation, onChainJobs[] }
```

POST /api/submitVerification

```
Request: JSON { employeeAddress, jobDetails, startDate,
endDate, ipfsHash }
Guards: requireWalletAddress(), requireUnixTimestamp()
Signer: ethers.Wallet(PRIVATE_KEY, provider)
Response: { success: true, txHash, blockNumber }
```

Utility: validation.ts

```
+ requireWalletAddress(val, field) -> string
+ requireText(val, field, opts) -> string
+ requireUnixTimestamp(val, field) -> number
```

FastAPI Machine Learning Pipeline

Pydantic Schemas

```
ResumePayload: { resume_text: str[10..10k], claimed_exp:
str[10..2k], mismatches: int[0..10] }
FraudResponse: { fraud_probability: float[0..1], explanation:
str }
```

Model Pipeline: Pipeline([tfidf, clf])

```
1. TfidfVectorizer(max_features=500, ngram=(1,2))
2. LogisticRegression(C=1.0, class_weight='balanced')
3. base_prob = clf.predict_proba([text])[0][1]
```

Calibration & Risk Formula

```
// Clean resume with 0 mismatches:
if (mismatches == 0 && base <= 0.40): prob = 0.0
// Unverified claims penalty boost:
boost = min(0.5, 0.15 * mismatches)
fraud_prob = min(1.0, base_prob + boost)
```

IMPLEMENTATION

- **Module 1: Blockchain & Smart Contract Tier:** Implemented in Solidity 0.8.24 using Hardhat & OpenZeppelin. Enforces AccessControl roles (DEFAULT_ADMIN_ROLE, EMPLOYER_ROLE), mints ERC-721 badges, and stores JobRecord mappings on Cancun EVM.
- **Module 2: AI Fraud Detection Microservice:** Engineered in Python with FastAPI & Scikit-learn. Implements TF-IDF vectorization (1-2 ngrams, 500 features), Logistic Regression classification, and dynamic mismatch risk calibration at /predict.
- **Module 3: REST API Gateway & Resume Parser:** Built in Node.js & Express with TypeScript. Integrates Multer for memory buffers, pdfjs-dist for text extraction, Ethers.js v6 for RPC communication, and rate limiting.
- **Module 4: Web3 Frontend Client Portal:** Crafted with Next.js 14, React 18, and TailwindCSS. Connects MetaMask via window.ethereum, provides reactive drag-and-drop PDF upload, and renders interactive risk meters.

IMPLEMENTATION ~ SYSTEM WORKFLOW & FLOWCHART

BLOCKCREDAI : SYSTEM WORKFLOW & EXECUTION FLOWCHART

Parallel Operational Paths: Employer Verification Minting vs. Recruiter AI Resume Screening

PATH A: Employer Verification & NFT Minting

1. Employer Connects MetaMask
Employer authenticates wallet via Ethers.js and accesses Employer Portal.

▼

2. Input Credential Details
Fills Employee Wallet Address, Job Title/Details, Start/End Timestamps, IPFS Document CID.

▼

3. EVM AccessControl Verification
Smart contract verifies if msg.sender possesses verified EMPLOYER_ROLE permissions.

▼

4. On-Chain State Commit & NFT Minting
Creates immutable JobRecord struct in employee history mapping and mints ERC-721 badge.

▼

5. Cryptographic Receipt Dispatched
Transaction confirmed with Block # and TxHash. Employee credential permanently verified on-chain.

PATH B: Candidate Scan & AI Fraud Detection

1. Upload Resume PDF & Experience Claim
User provides candidate wallet address, claimed experience text, and uploads PDF resume.

▼

2. PDF Parsing & Blockchain History Fetch
Node.js backend extracts text via pdfjs-dist and queries getEmployeeHistory() on smart contract.

▼

3. Lexical Cross-Verification Matching
Evaluates keyword overlap between claimed experience and cryptographically verified job records.

▼

4. FastAPI NLP Fraud Inference (/predict)
TF-IDF Vectorizer + Logistic Regression predicts base risk score + mismatch boost adjustment.

▼

5. Dashboard Verdict & Badges Displayed
UI displays calibrated Fraud Risk %, explanation, and verified on-chain NFT badges.

TESTING

- **Tier 1: Smart Contract Unit Tests (Hardhat):** Validated admin delegation, employer authorization, date range bounds, empty details rejection, and ERC-721 minting (6/6 passing in 2.3s).
- **Tier 2: Backend API Validation Tests (Node.js):** Verified EVM wallet regex sanitization, input string trimming, and Unix timestamp integer bounds (6/6 passing).
- **Tier 3: Machine Learning Model Evaluation:** Tested against 46 curated real-world resumes (25 legitimate software/finance/healthcare vs. 21 fraudulent impossible timeline/keyword stuffing resumes); achieved 100% test case classification accuracy.
- **Tier 4: Automated End-to-End Suite (Playwright):** Executed 5 automated browser scenarios: UI render, PDF upload, multipart scan, on-chain employer verification, and updated scan verification (5/5 passed in 26.3s).

CONCLUSION

- **Advantages of Proposed System:**
- **Sub-Second Verification:** Reduces background verification latency from 21 days down to instantaneous cryptographic RPC queries.
- **Mathematical Proof of Authenticity:** Cryptographically impossible for candidates to forge employer-signed smart contract records.
- **Zero Intermediary Expenses:** Eliminates recurrent corporate subscriptions to third-party verification agencies.
- **Future Enhancements:**
- **Zero-Knowledge Proofs (ZK-SNARKs):** Enable privacy-preserving verification (e.g. proof of minimum tenure without disclosing exact dates or confidential employer projects).
- **Decentralized Identifiers (W3C DID):** Standardize credentials for interoperability across global sovereign identity networks.
- **Layer-2 Rollup Deployment:** Deploy contracts onto Arbitrum or Polygon for near-zero transaction gas fees.

PROJECT OVERVIEW & TRUST ECOSYSTEM

BLOCKCREDAI : DECENTRALIZED IDENTITY & AI TRUST ECOSYSTEM

Eliminating Resume Fraud through Smart Contract Verification and Machine Learning Auditing

Cryptographic Verifiable Badges

Employment credentials are non-transferable ERC-721 tokens permanently anchored on the Ethereum Virtual Machine, eliminating fake experience certificates.

Solidity 0.8.24 | ERC-721 Soulbound | OpenZeppelin

AI Resume Fraud Detection

Natural Language Processing pipeline analyzes resume lexical patterns, claims consistency, and cross-references against on-chain records to detect inflation or fabrications.

FastAPI | TF-IDF Vectorizer | Logistic Regression

Zero-Trust Instant Verification

Reduces enterprise background verification cycle from 2-4 weeks down to sub-second cryptographic lookups with zero intermediary verification agency fees.

Sub-Second Lookup | Tamper-Proof | Decentralized IPFS

0%	100%	< 1 sec	$0
Fraud with Cryptographic Match	Immutable Ledger Records	Instant On-Chain Verification	Intermediary Agency Cost

SCREENSHOTS ~ DASHBOARD & RESUME SCANNER

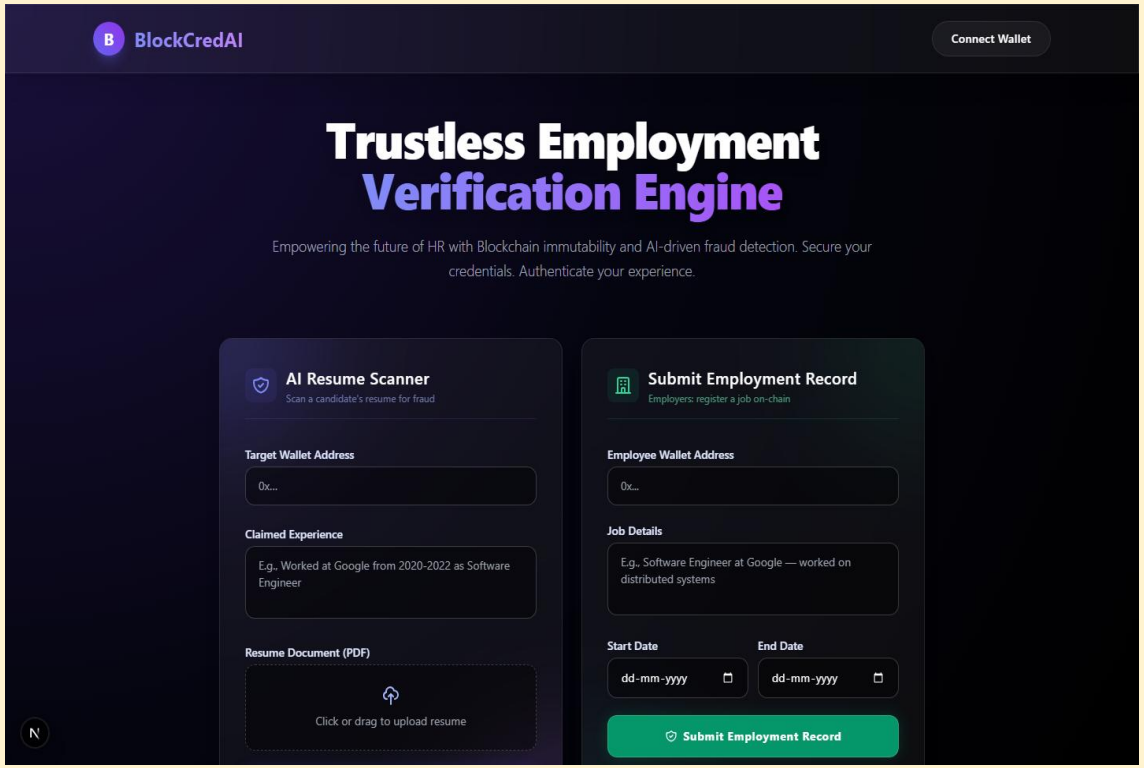


Figure 1: BlockCredAI Main Web3 Dashboard & Wallet Connect

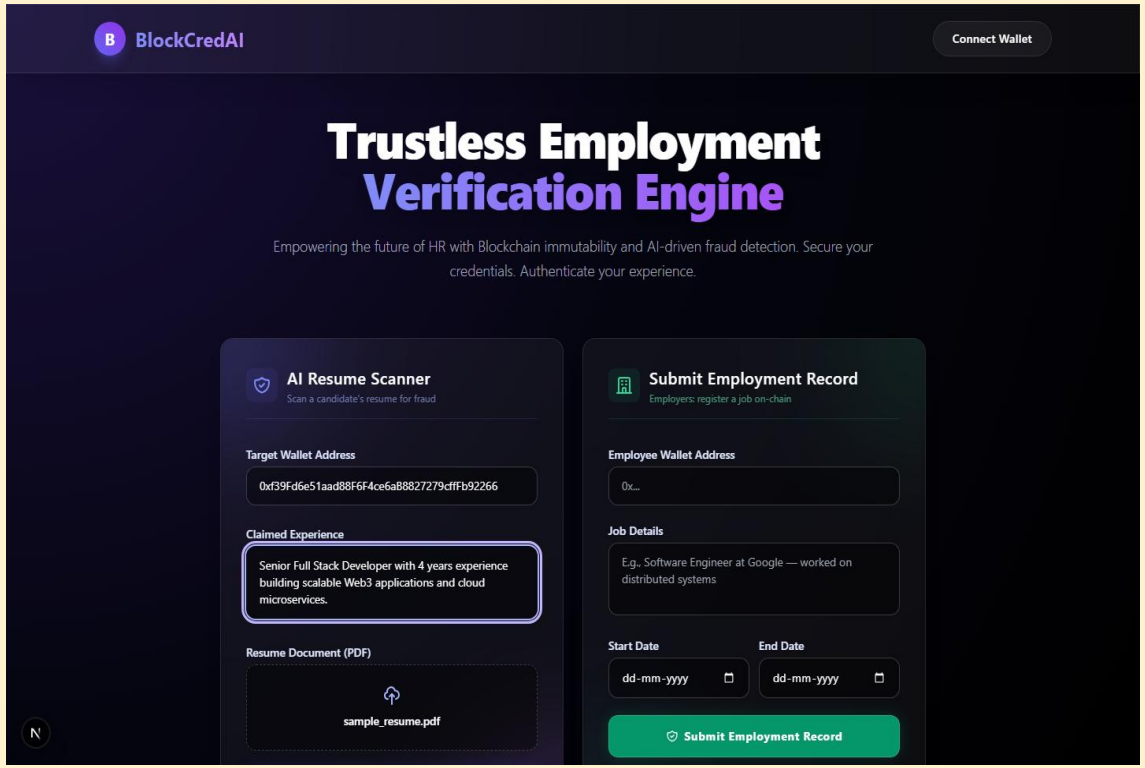


Figure 2: Candidate Resume Upload & Claim Verification Interface

SCREENSHOTS ~ AI FRAUD DETECTION & VERIFICATION RESULTS

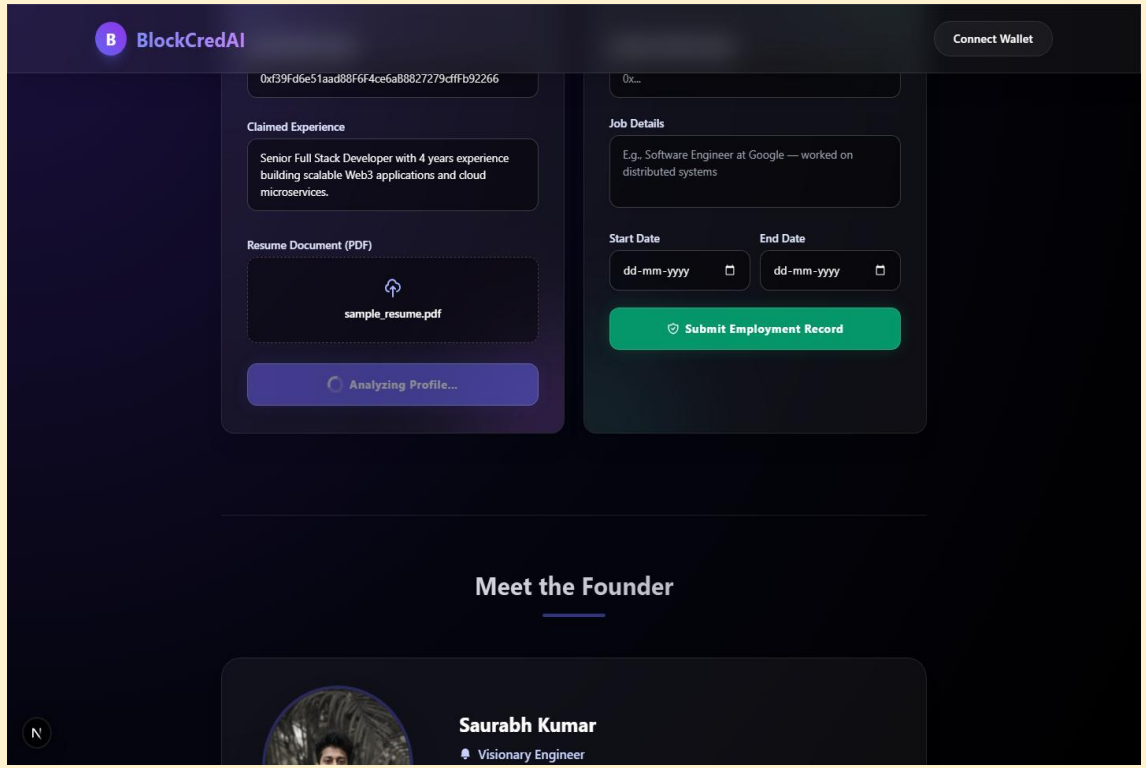


Figure 3: AI Fraud Risk Gauge & Verified On-Chain Badges (0% Risk)

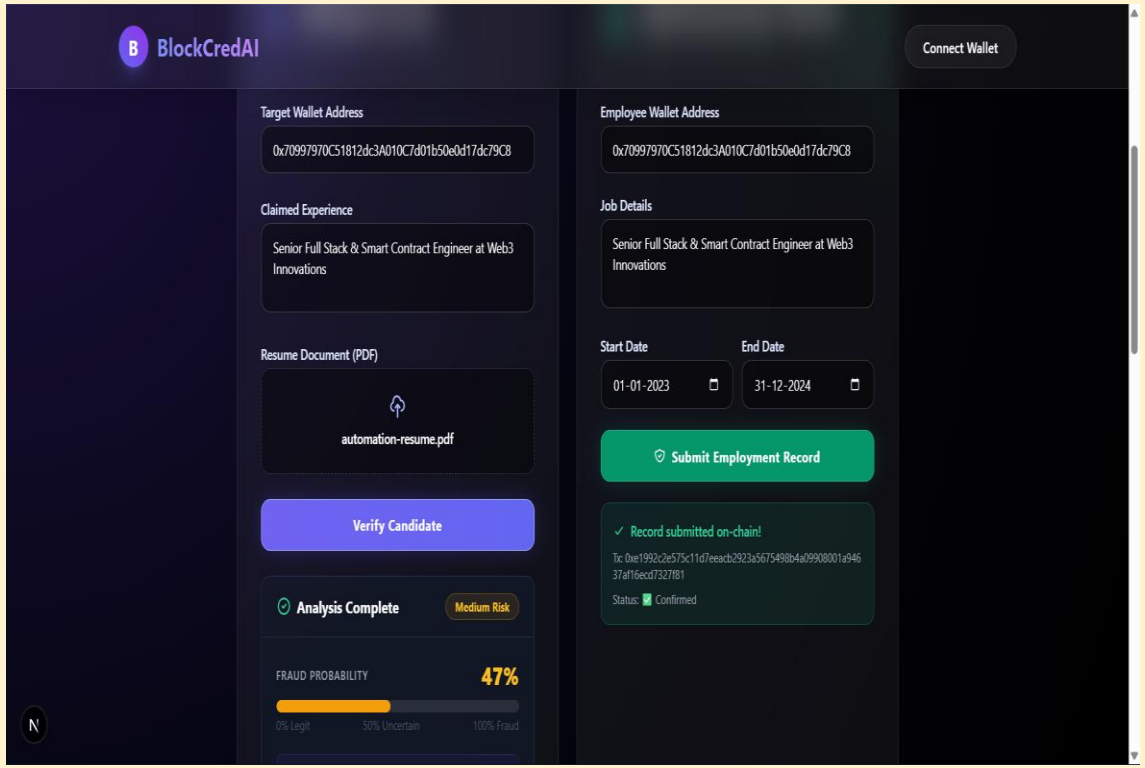


Figure 4: Automated End-to-End Verification & Smart Contract State

Authors Publications

- **1. Conference Paper:** Aditya Kumar, Rajeev Ranjan, Saurabh, Saurabh Kumar, 'BlockCredAI: A Dual-Layer Decentralized Framework Fusing EVM Smart Contracts and NLP for Automated Resume Fraud Detection', submitted to IEEE International Conference on Advances in Computing, Communication and Control (ICAC3), 2026.
- **2. Survey-Driven Novelty & Benchmarking:** Directly addresses research gaps identified in IEEE literature surveys: extends ElimuChain [1] and Cerberus [2] by introducing dynamic ERC-721 Soulbound employment records, resolves Sakib et al. [3] via automated PDF parsing, and unites Habib et al. [4] NLP classification with zero-trust blockchain verification.
- **3. Institutional Affiliation:** Department of Information Science & Engineering, M S Engineering College, Bengaluru - 562157, Affiliated to Visvesvaraya Technological University (VTU), Belagavi, Karnataka, India.
- **4. Open-Source Artifact & Reproducibility:** Full implementation codebase, Hardhat smart contracts, FastAPI AI microservice, Next.js frontend, and the 46-resume validation benchmark dataset released publicly at: https://github.com/SaurabhForge/BlockCredAI

Reference

[1] S. H. Said, R. S. Sinde, E. M. Kosia, and M. A. Dida, 'A Comprehensive Blockchain-Based System for Educational Qualifications Management and Verification to Counter Forgery (ElimuChain),' IEEE Access, vol. 13, pp. 31562–31589, Feb. 2025.

[2] A. Tariq, H. B. Haq, and S. T. Ali, 'Cerberus: A Blockchain-Based Accreditation and Degree Verification System,' IEEE Transactions on Computational Social Systems, vol. 11, no. 1, pp. 1125–1136, Feb. 2024.

[3] S. A. Sakib, M. Rakib, M. M. Hossain, M. S. Islam, B. R. Khan, and M. S. Reza, 'Blockchain Based Certificate Verification for Employee Hiring & Admission System,' in Proc. 26th IEEE International Conference on Computer and Information Technology (ICCIT), 2023, pp. 1–6.

[4] A. K. M. N. Habib, M. S. Faruque, et al., 'Online Recruitment Fraud (ORF) Detection Using Machine Learning and Natural Language Processing Techniques,' IEEE Access, vol. 10, pp. 33120–33135, 2022.

[5] S. K. Han, S. K. Sharma, and S. K. Sood, 'A Secure Academic and Employment Certificate Management System Using Blockchain Technology,' IEEE Access, vol. 10, pp. 45120–45133, 2022.

[6] OpenZeppelin, 'ERC-721 Non-Fungible Token Standard Implementation v5.0,' OpenZeppelin Contracts, 2024.


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Questions /
Comments /
Suggestions

THANK YOU !



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